

Baggage Operations Dashboard

Overview

A comprehensive operational dashboard tracking baggage processing through security scanning systems at Manchester Airport. Monitors throughput, scan accuracy, machine performance, recirculation rates, and secondary screening efficiency across three terminals.

Key Metrics

Metric	Value	Definition
Unique Bags Processed	847,139	Total distinct bags scanned
Scan Results	Cleared: 85% Rejected: 10% Unclear: 5%	Outcome distribution (Cleared/Rejected/Unclear)
Peak Throughput (15 min)	23,793 bags	
Peak Hourly Volume	83,854 bags	Busiest hour (08:00–09:00)
Avg Decision Time L2	49.4 seconds	Time for secondary screening decision
Recirculation Rate	6.3%	Bags requiring re-scan (quality control)

Dashboard Pages

Overview & Scan Result Breakdown: Headline KPIs, scan outcome distribution, time-based peaks, and daily throughput trend. Filters by operational hours (shift vs. non-shift).

Machine Performance: Individual scanner utilisation (Clusters A, B; Machines 1–8). Shows % scans per machine and days in operation, identifying under/over-utilised equipment.

Level 2 & Recirculation: Secondary screening metrics. Tracks recirculation % by machine and daily trends. Identifies machines requiring more thorough first-pass scanning.

L2 Bags & Decision Time: Secondary scan outcomes (Explosives, None, Timeout, No decision). Average cycle time and bag count by outcome reason.

Use Cases

- **Capacity Planning:** Identify peak periods and machine utilisation trends to optimise scheduling.
- **Quality Monitoring:** Track recirculation rates to spot machines requiring maintenance or calibration.
- **Performance Benchmarking:** Compare machine performance; identify outliers.
- **Incident Investigation:** Drill into specific dates/shifts to understand scan failures or delays.

Technical Stack

Component	Technology
Visualisation	Power BI Desktop / Service
Data Source	Airport baggage system database
Refresh Frequency	Daily (23/10/25 last update)
Key Filters	Date, Operational Hours (shift classification)